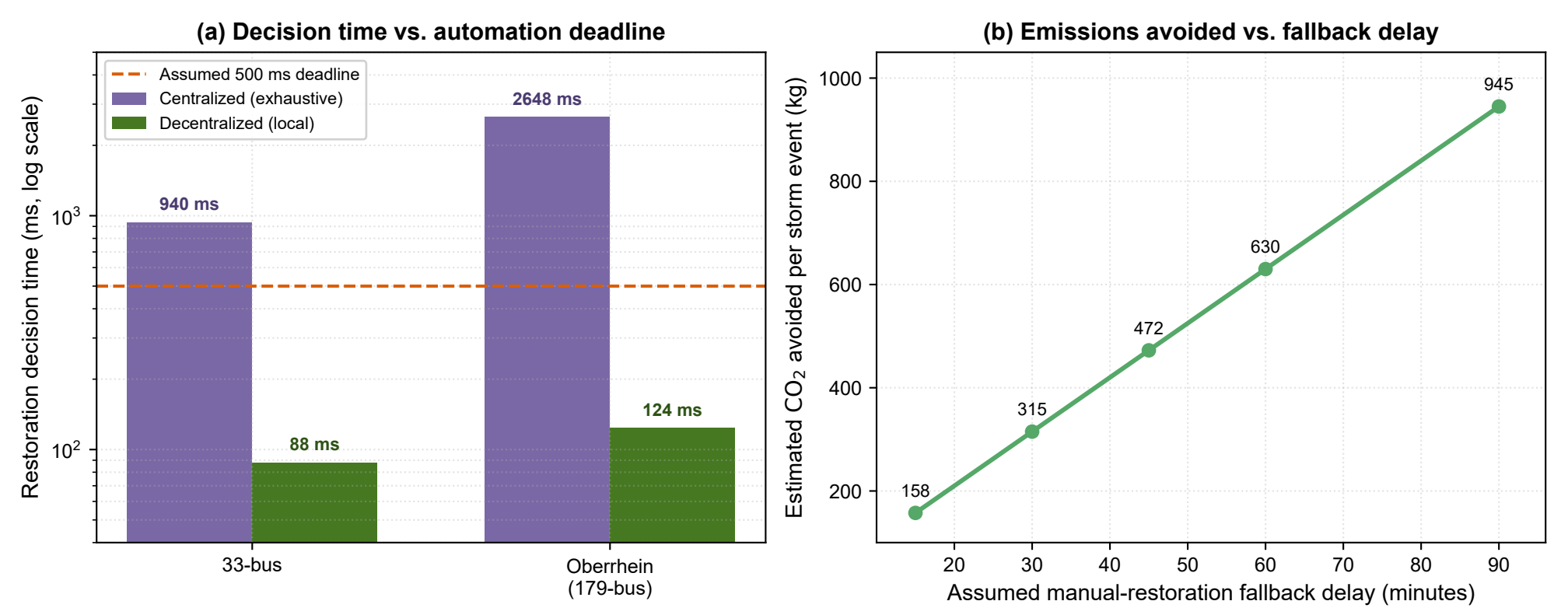
(a) Decision time vs. automation deadline
Assumed 500 ms deadline
Centralized (exhaustive)
Decentralized (local)
940 ms
88 ms
2648 ms
124 ms
Restoration decision time (ms, log scale)
10^3
10^2
33-bus
Oberrhein
(179-bus)
(b) Emissions avoided vs. fallback delay
Estimated CO_2 avoided per storm event (kg)
1000
800
600
400
200
158
315
472
630
945
20
30
40
50
60
70
80
90
Assumed manual-restoration fallback delay (minutes)